Relationship among Total, Average and Marginal Products

Stages	Total Product	Marginal Product	Average Product
Stage I	Initially it increases at an increasing rate and then increases at a decreasing rate	At the beginning it increases, then reaches a maximum and starts to decrease	At the first instant it increases, then attains maximum
Stage II	It continues to increase at a diminishing rate and reaches maximum.	It continues to diminish and becomes equal to zero	It is equal to MP and then begins to diminish
Stage III	It diminishes	It becomes negative	It continues to diminish but always greater than zero (positive)

3.5 Laws of Returns to Scale

In the long- run, there is no fixed factor; all factors are variable. The laws of returns to scale explain the relationship between output and the scale of inputs in the long-run when all the inputs are increased in the same proportion.

Assumptions

Laws of Returns to Scale are based on the following assumptions.

- All the factors of production (such as land, labour and capital) are variable but organization is fixed.
- There is no change in technology.
- There is perfect competition in the market.
- Outputs or returns are measured in physical quantities.

Three Phases of Returns to Scale

(1) Increasing Returns to Scale:

In this case if all inputs are increased by one percent, output increase by more than one percent.

(2) Constant Returns to Scale:

In this case if all inputs are increased by one percent, output increases exactly by one percent.

(3) Diminishing Returns to Scale:

In this case if all inputs are increased by one percent, output increases by less than one percent.

Diagrammatic Illustration

The three laws of returns to scale can be explained with the help of the diagram below.

In the diagram 3.2, the movement from point **a** to point **b** represents

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